

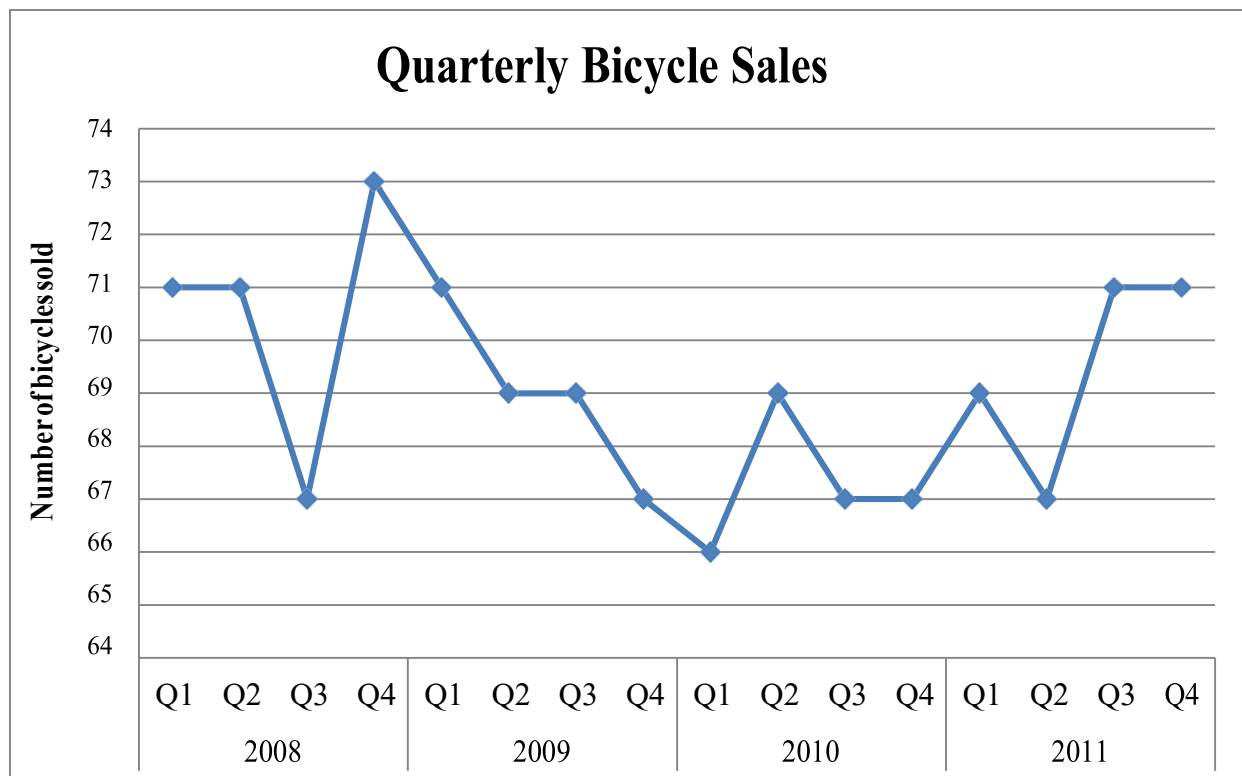
Week 10 Tutorial – Time Series

Question 1

Consider the quarterly bicycle sales for a sports shop in Table 1.1

Table 1.1: Quarterly bicycle sales

Year	2008				2009				2010				2011			
Quarter	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Bicycles	71	71	67	73	71	69	69	67	66	69	67	67	69	67	71	71



- Identify the time series components visible in the plotted series
- When smoothing via moving averages, justify why (or why not) $k = 4$ would (or would not) be appropriate.
- Two smoothing techniques were used to smooth the data, (centred) moving averages (with $k = 4$) and simple exponential smoothing (with a smoothing constant of 0.25). Fill in the missing values, A to D. Round answers off to THREE decimal places

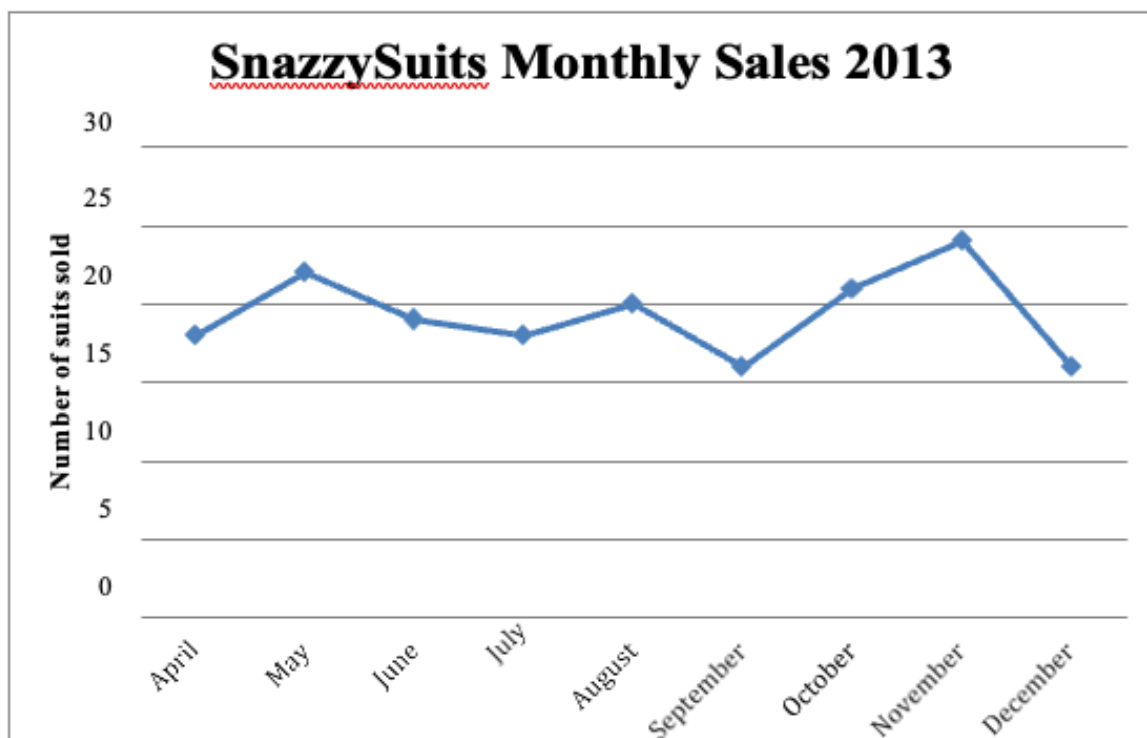
Year	Quarter	Bicycles	CMA($k = 4$)	Exp ($\alpha = 0.25$)
2008	Q1	71		C
	Q2	71		71.000
	Q3	67	A	70.000
	Q4	73	70.250	70.750
2009	Q1	71	70.250	70.813
	Q2	69	69.750	70.359
	Q3	69	68.375	70.020
	Q4	67	67.750	D
2010	Q1	66	67.500	68.448
	Q2	69	67.250	68.586
	Q3	67	67.625	68.190
	Q4	67	67.750	67.892
2011	Q1	69	B	68.169
	Q2	67	69.000	67.877
	Q3	71		68.658
	Q4	71		69.243

Question 2

Harvey and Mike are the proud owners of a new store selling suits, *SnazzySuit*. They opened for business on the 1st of April 2013, and are concerned about the future prospects of their store. They provided a record of their monthly sales up to the end of 2013 to Rachel, a STA2020 student and requested that she analyze the data and inform them how many suits they can expect to sell in January and February 2014. The data are presented in the table below.

Month	April	May	June	July	August	September	October	November	December
# suits sold	18	22	19	18	20	16	21	24	16

Monthly sales of *SnazzySuit* Store



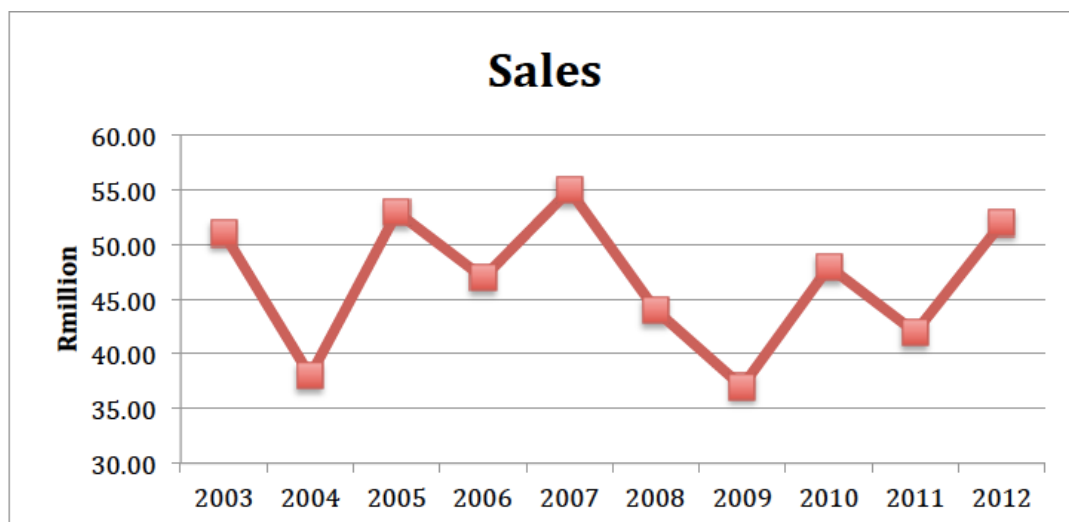
- Comment on what time series components are visible in the data.
- Name an advantage that exponential smoothing has over moving averages.
- Smooth the series using a simple exponential smoothing model with $\alpha = 0.4$

Question 3

Consider the annual sales (in R million) of Clearview Water Holdings Inc.

Table 1.1 Annual Sales of Clearview Water Holdings Inc.

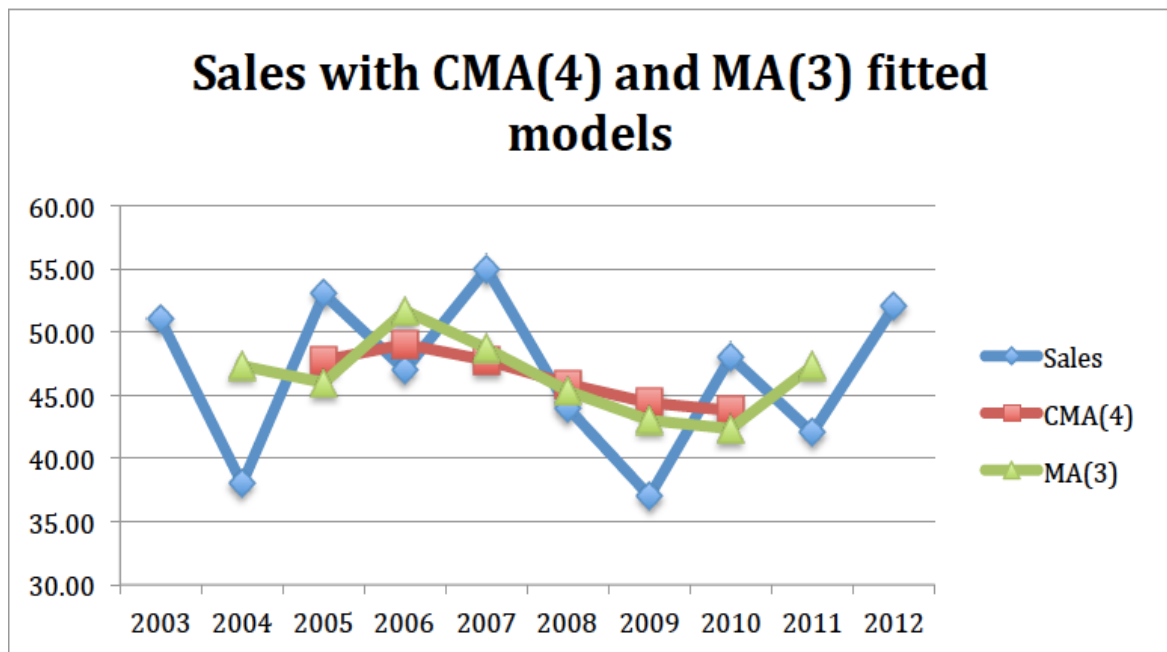
t	Year	Sales	CMA(4)	Exp(0.2)
1	2003	51		D
2	2004	38		48.4
3	2005	53	A	49.32
4	2006	47	49	48.86
5	2007	55	47.75	E
6	2008	44	B	48.87
7	2009	37	44.38	46.49
8	2010	48	C	46.8
9	2011	42		F
10	2012	52		47.07



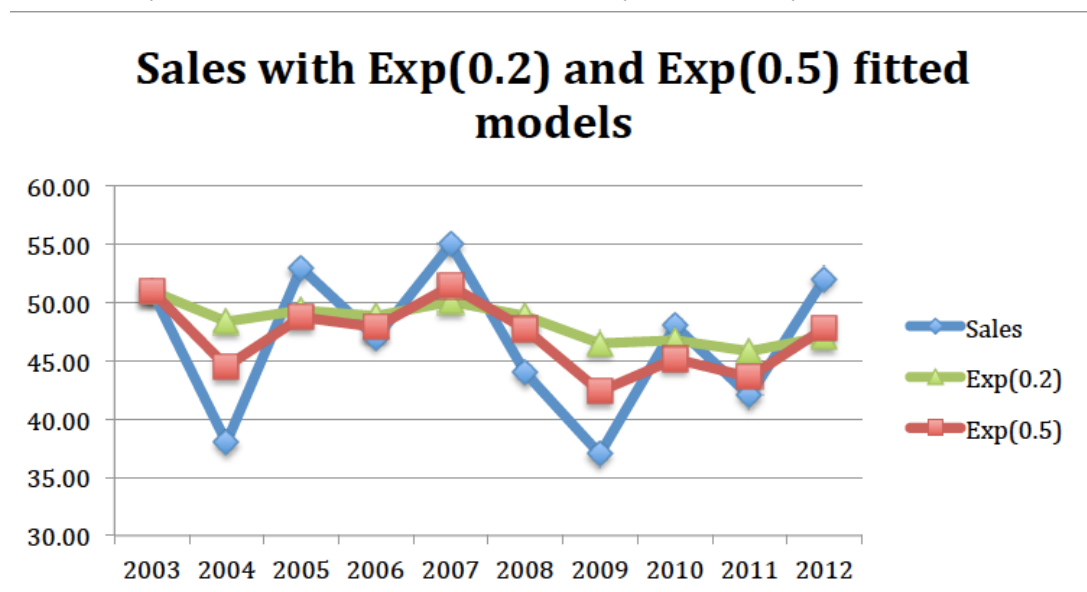
- Identify any components visible in the time series.
- Calculate the missing values A – F in Table 1.1. Round answers to 2 decimal places.
- Compute a MA(3) and an Exp(0.5) smoothed series of the data.

Consider two new plots of the time series:

- i. A plot with the original data and the CMA(4) and MA(3) series



- ii. A plot with the original data and the Exp(0.2) and Exp (0.5) series

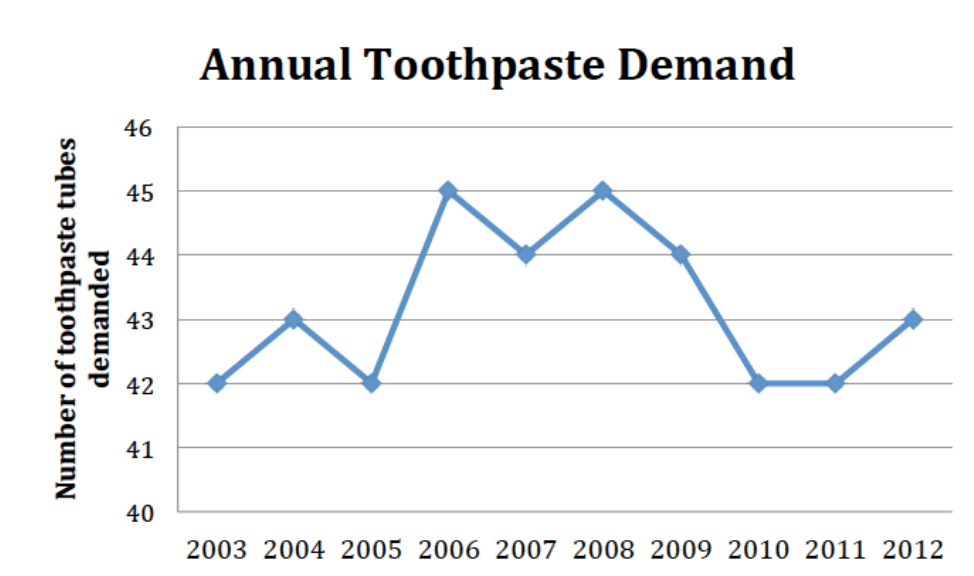


- d) For your plot in (d) (i): What has the effect been of choosing a smaller value of k ?
- e) For your plot in (d) (ii): What has the effect been of choosing a larger value of α ?
- f) What are the advantages and disadvantages of the moving average smoothing technique?
- g) What are the advantages of exponential smoothing?
- h) Are the smoothing models fitted in this question suitable to forecast future values of the time series? Justify your answer.

Question 4

Consider the demand of toothpaste tubes (in thousands) at a particular retailer:

2003	2004	2005	2006	2007	2008	2009	2010	2011	2012
42	43	42	45	44	45	44	42	42	43



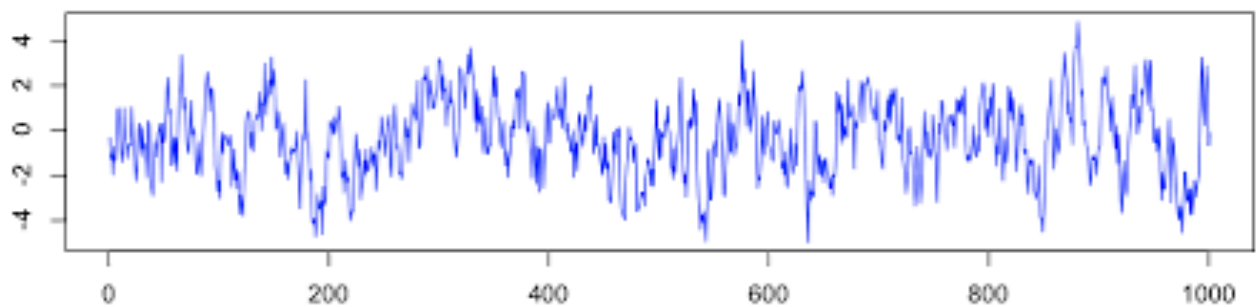
- Would exponential smoothing be a suitable forecasting method to use for the stated time series? Justify your answer.
- Why would exponential smoothing be preferred over moving averages?
- Assuming that exponential smoothing is appropriate, apply the technique with $\alpha = 0.2$.
- What would the effect of a larger α have been?
- Predict the demand for the next time period, 2013. Round your answer to the closest thousand.

Question 5

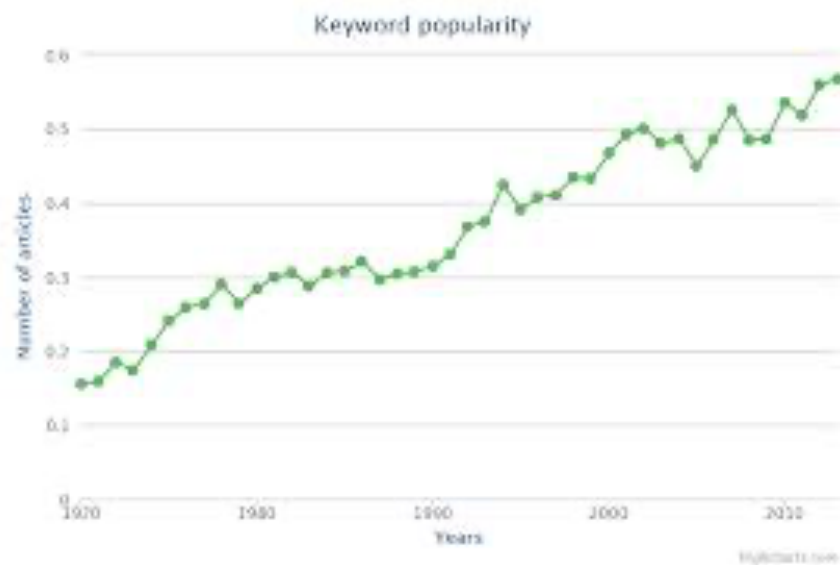
For each of the following examples:

- State the components present in the time series.
- Indicate whether the time series is stationary or not.
- Indicate whether and what transformations may be applicable to transform the time series into a stationary series.
- State and justify what would be the most appropriate simple forecasting method for the time series

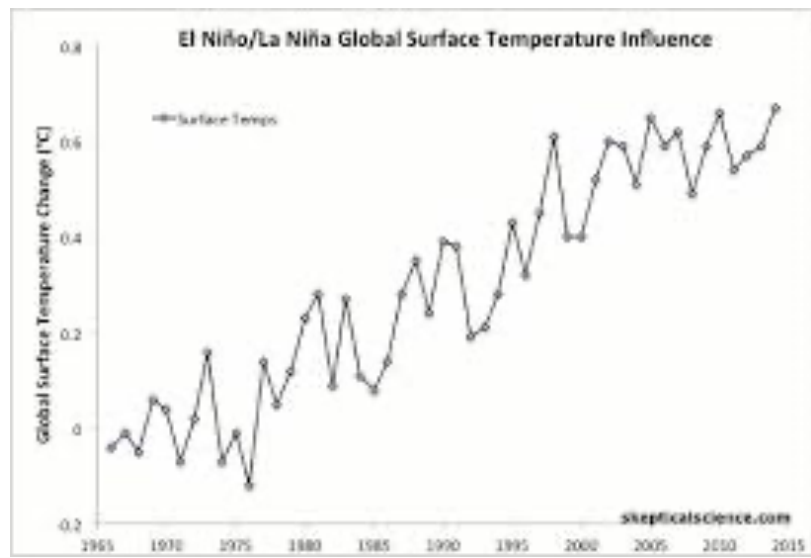
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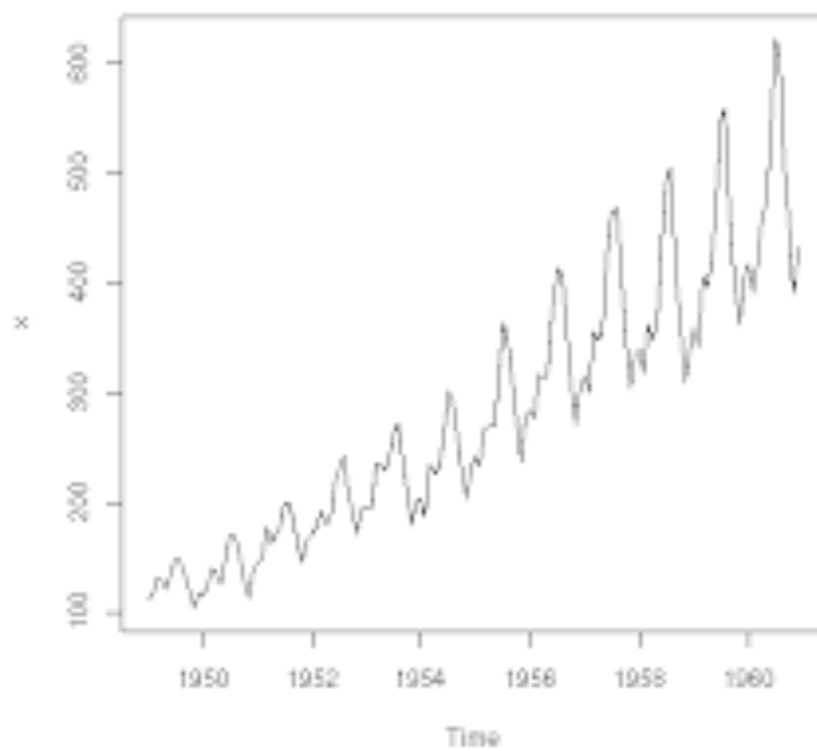
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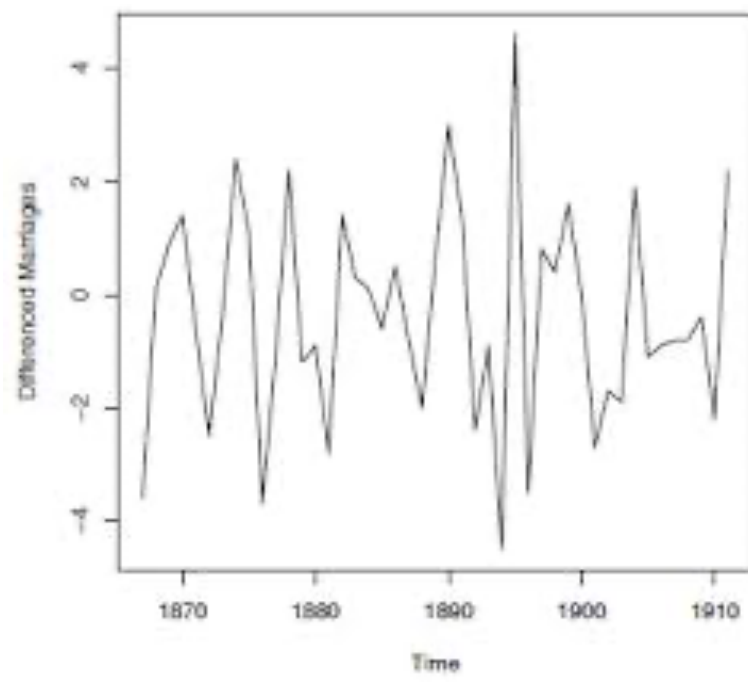
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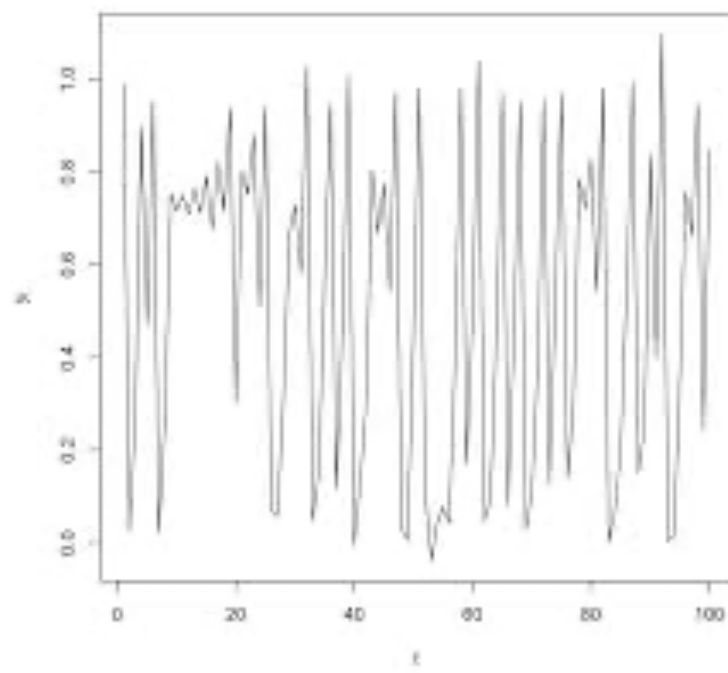
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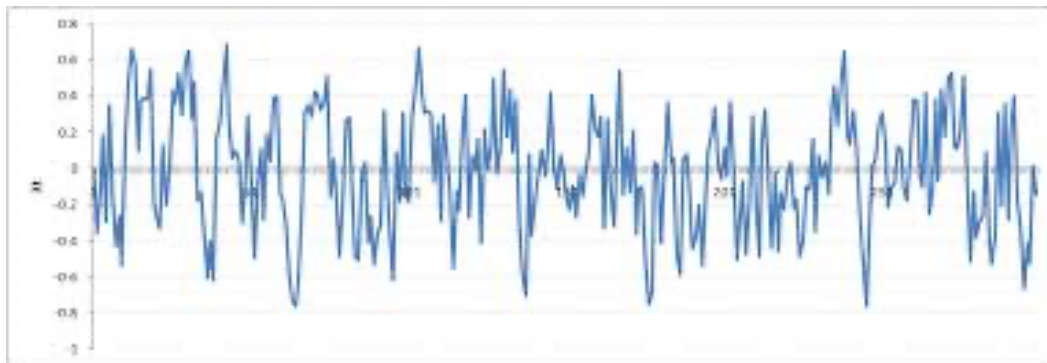
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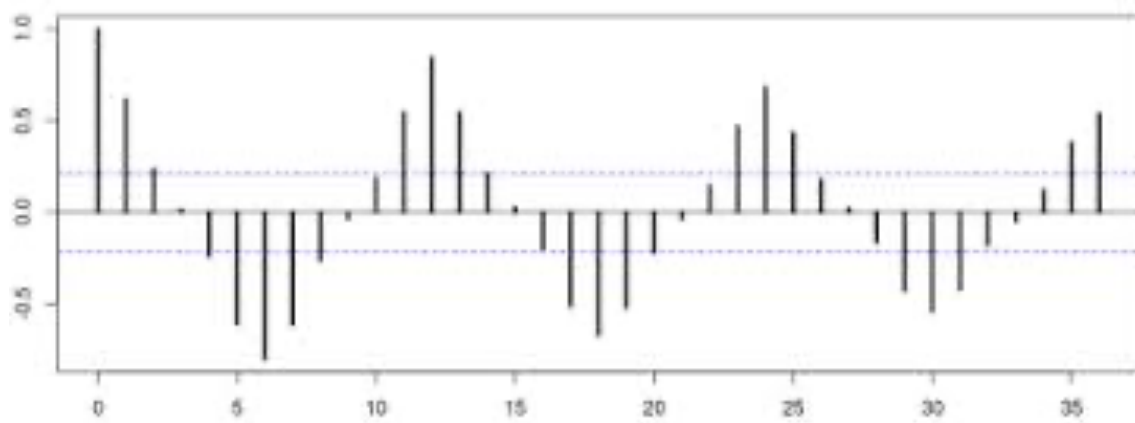
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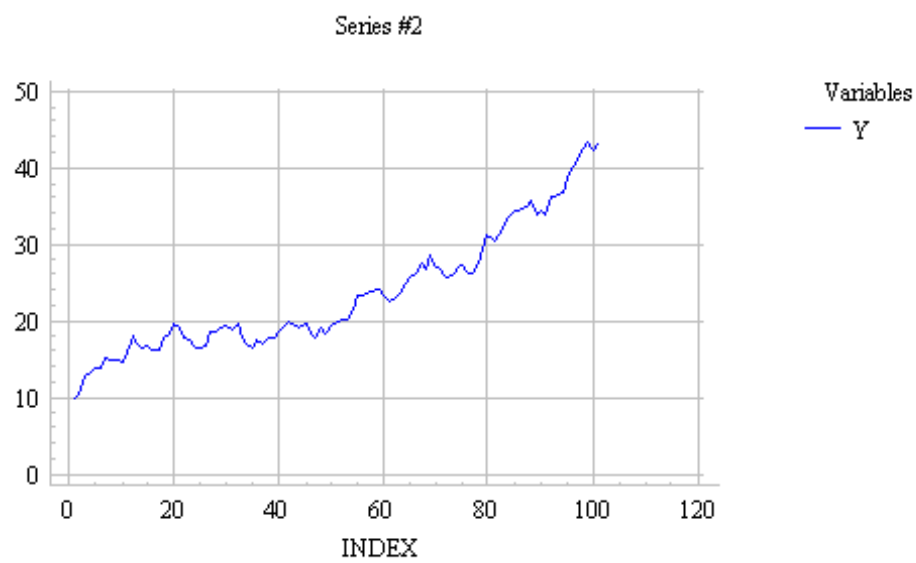
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Question 6

Consider the Google stock price data presented in the time series plot and table below.



Most recent 10 days stock price

Date	Stock price
2017/02/04	802.32
2017/02/05	796.79
2017/02/06	795.70
2017/02/07	798.53
2017/02/08	801.49
2017/02/09	801.34
2017/02/10	806.97
2017/02/11	808.38
2017/02/12	809.56
2017/02/13	813.67

A drift model was fitted to the above data, with a summary of the fitted model below:

Forecast method: Random walk with drift

Model Information:

Call: rwf(y = goog, h = 5, drift = T)

Drift: 0.4213 (se 0.2762)

Residual sd: 8.7285

- a) Compute an 80% prediction interval for the forecasted stock price for 14th February 2017
- b) Compute a 90% prediction interval for the forecasted stock price for 18th February 2017
- c) Compute a 60% prediction interval for the forecasted stock price for 25th February 2017

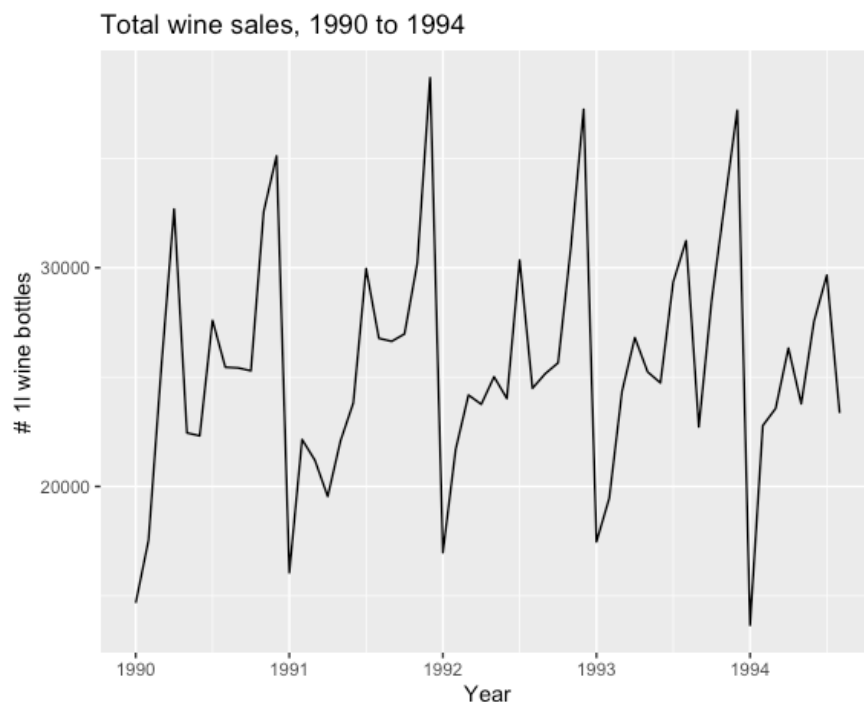
Three models were fitted to the entire dataset, and their forecasts for the next 5 days, along with the actual stock price values, are included in the table below.

Date	Stock price	Mod 1	Mod 2	Mod 3
2017/03/14	820.45	813.67	814.09	814.08
2017/03/15	818.98	813.67	814.51	814.49
2017/03/16	824.16	813.67	814.93	814.90
2017/03/17	828.07	813.67	815.36	815.31
2017/03/18	831.66	813.67	815.78	815.72

- d) Explain what both MAD and MSE measure, and how they can be used to assess the fit of a model
- e) Compute the MAD for the above three models
- f) Based on the computed MAD values, which model would you choose as most appropriate to forecast future Google stock price values? Justify your answer.

Question 7

Consider the Australian total monthly wine sales by wine makers in bottles ≤ 1 litre, from Jan 1990 – Aug 1994 below.



- What components are evident from the time series plot?
- Do you expect autocorrelation to be present in the time series? If yes, at what lag?
- Is this time series stationary? Justify your answer
- Are any transformations needed? If yes, name and describe the transformations you would apply.
- Given the data below on the autocorrelation coefficients, test for the presence of autocorrelation at lag 3. State the appropriate test to use, the hypotheses, calculate the test statistic and state your decision and conclusion based on the p-value of the test statistic.

Lag	Autocorrelation
1	0.091
2	-0.225
3	-0.116
4	0.089
5	-0.06
6	-0.345
7	0.058
8	0.101
9	-0.154
10	-0.209
11	0.047
12	0.61